13.3inch e-Paper HAT+ (E) Manual

From Waveshare Wiki
Jump to: navigation, search

Overview

Parameters

13.3inch e-Paper (E)
Raw display only

(https://www.waveshare.com/13.3inch-e-paper-hat-plus-e.htm?sku=29353)

1600×1200, RPi/SPI, E6 Full Color e-Paper

13.3inch e-Paper HAT+ (E)
With Driver HAT

(https://www.waveshare.com/13.3inch-e-paper-hat-plus-e.htm?sku=29355)

1600×1200, RPi/SPI, E6 Full Color e-Paper

Size	13.3inch
Driver Board Size	65.00mm × 30.50mm
Display Size	270.40mm × 202.80mm
Outline Dimensions	284.70mm × 208.80mm × 0.85mm

Operating Voltage	3.3V / 5V (IO level voltage should be the same as the supply voltage)
Communication Interface	SPI
Dot Pitch	0.169mm × 0.169mm
Resolution	1600 × 1200
Display Color	E6 Full Color
Grayscale	2
Refresh Time	19s
Total Refresh Power	<0.5W（typ.)
Operating Temperature	0 ~ 40 ℃
Storage Temperature	-25 ~ 60 ℃

- Refresh time: The refresh time is based on experimental test data, and there may be errors in the actual refresh time. The actual results shall prevail. There will be a flickering effect during the global refresh process, which is a normal phenomenon.
- Refresh power consumption: The power consumption data is experimental test data, and the actual power consumption may have certain errors due to the presence of the driver board and actual usage. The actual results shall prevail.

- Refresh at low temperatures may cause a color cast, and it is necessary to stand still at 25°C for 6 hours before refreshing.

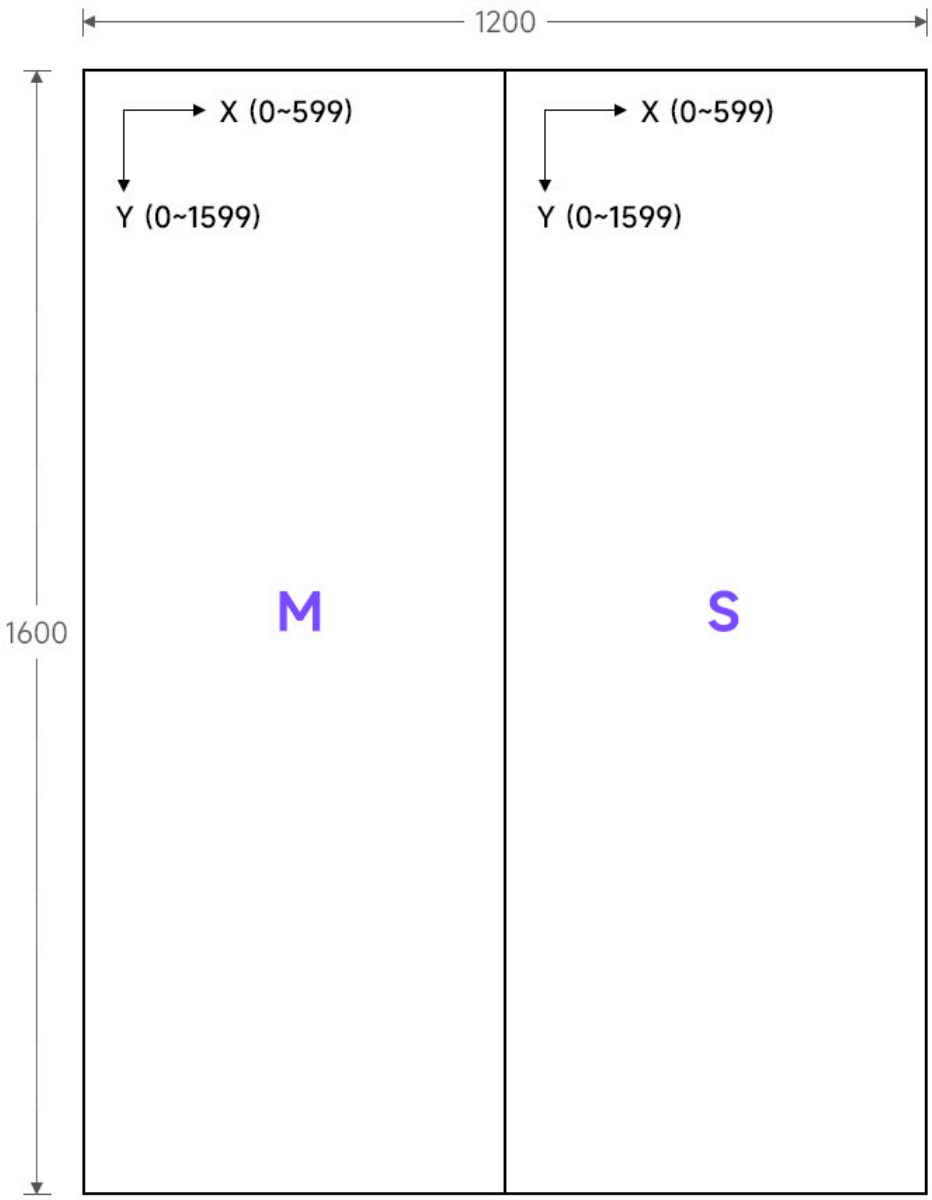
Communication Method [Expand]

Operating Principle [Expand]

Programming Principle [Expand]

Double-IC Programming Analysis [Collapse]

The 13.3inch e-Paper (E) is a dual-IC controlled e-Paper screen, in addition to the above programming principle, please note the following



(/wiki/File:13.3inch-e-Paper-E-IC.jpg)

- As shown in the figure above, each IC controls half of the screen, and the two CS pins are used to control the corresponding M area (master area) and S area (slave area)
- Except for CS pins, other control pins are shared

- Except for common commands (initialization command and refresh command in the demo), only one CS pin is allowed to be pulled low at a time
- When transmitting images, it should be noted that the display direction of the screen is horizontal display (X-axis direction), and after reaching the end point of a region, it will be displayed in the next line. Therefore, it is necessary to cut the picture into two pictures and transmit them separately, otherwise it will be displayed misalignment

Precautions

[Collapse]

1. Note that the screen cannot be powered on for a long time. When the screen does not refresh, the screen must be set to sleep mode or power off. Otherwise, the screen will remain in a high voltage state for a long time, which will damage the diaphragm and cannot be repaired.
2. When using the e-paper screen, it is recommended that the refresh interval is at least 180s, and refreshed at least every 24 hours. If the e-paper screen is not used for a long time, it should be whitened and stored. (Refer to the data sheet for specific storage environment requirements)
3. After the screen enters sleep mode, the image data sent will be ignored, and it can only be refreshed normally after re-initialization.
4. If you find that the data of the created image is displayed incorrectly on the screen, it is recommended to check whether the image size is set correctly, and try again by changing the width and height settings.
5. The working voltage requirement for an ink screen is 3.3V. If you purchase a raw display and need to work in a 5V environment when designing the circuit, it is recommended to perform level conversion processing.
6. The FPC cable of the screen is relatively fragile. Please note: do not bend the cable in the vertical direction of the screen to avoid the cable being torn; do not repeatedly bend the cable excessively to avoid the cable breaking; do not bend the cable toward the front of the screen to prevent the cable from being disconnected from the panel. During debugging and R&D, it is recommended to fix the wiring before use.
7. The ink screen screen is relatively fragile, so be careful to avoid falling, colliding, or pressing hard.
8. We recommend that customers first use the sample demo we provide and use the corresponding development board to test after getting the screen.

Picture Processing

[Collapse]

Picture production and conversion of multi-color e-Paper

Picture Production

[Expand]

Picture Data Conversion

[Expand]

Raspberry Pi

[Collapse]

Hardware connection

When connecting to the Raspberry Pi, you can directly plug the board into the 40PIN pin header of the Raspberry Pi, and be careful to align the pins.
If you choose to use an 10PIN cable to connect, please refer to the pin corresponding table below:

Raspberry Pi connection pin correspondence

e-Paper	Raspberry Pi	
	BCM2835 encoding	Board physical pin number
VCC	3.3V	3.3V
GND	GND	GND
DIN	MOSI	19
CLK	SCLK	23
CS_M	CE0	24
CS_S	CE1	26
DC	25	22
RST	17	11
BUSY	24	18

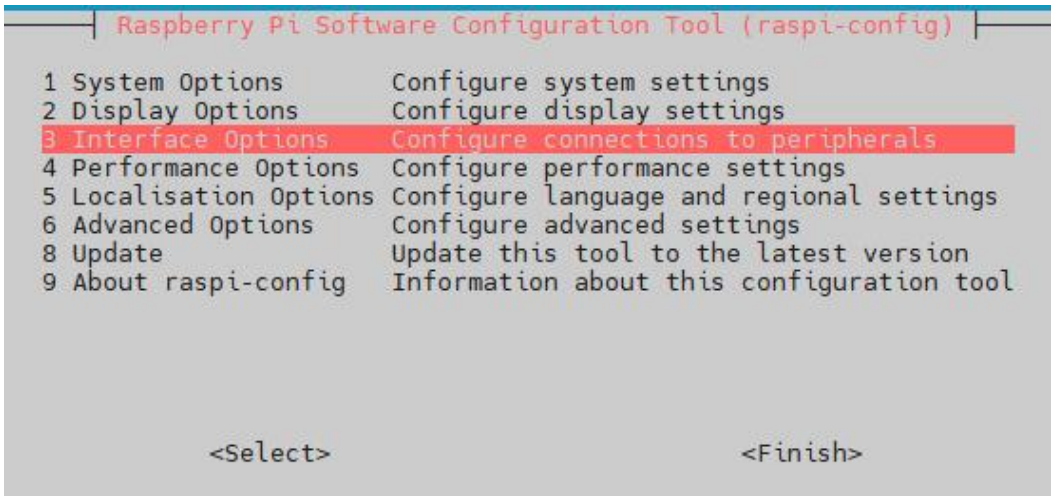
PWR	18	12

Raspberry Pi Preparation

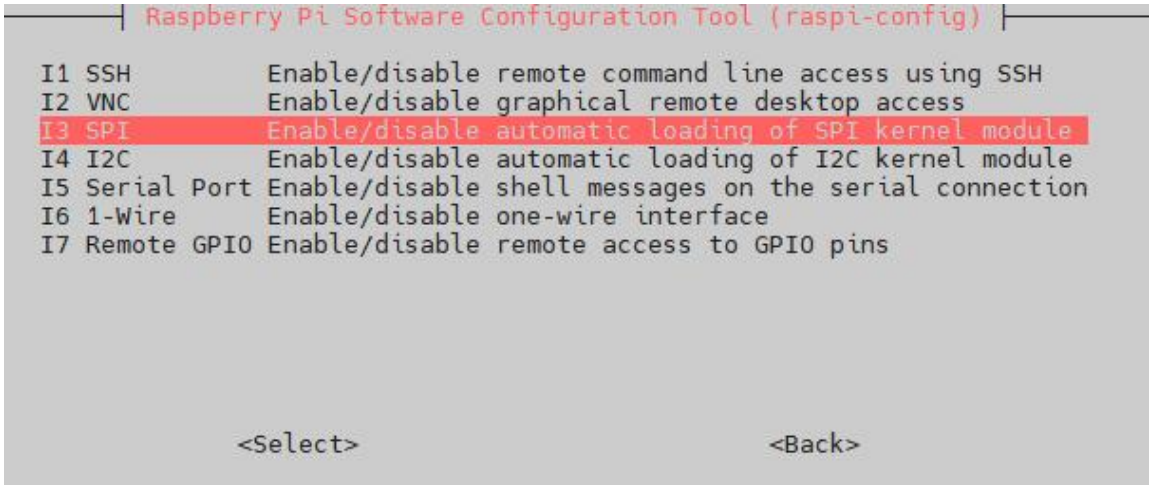
Disable SPI interface

- Open the Raspberry Pi terminal and enter the following command to enter the configuration interface:

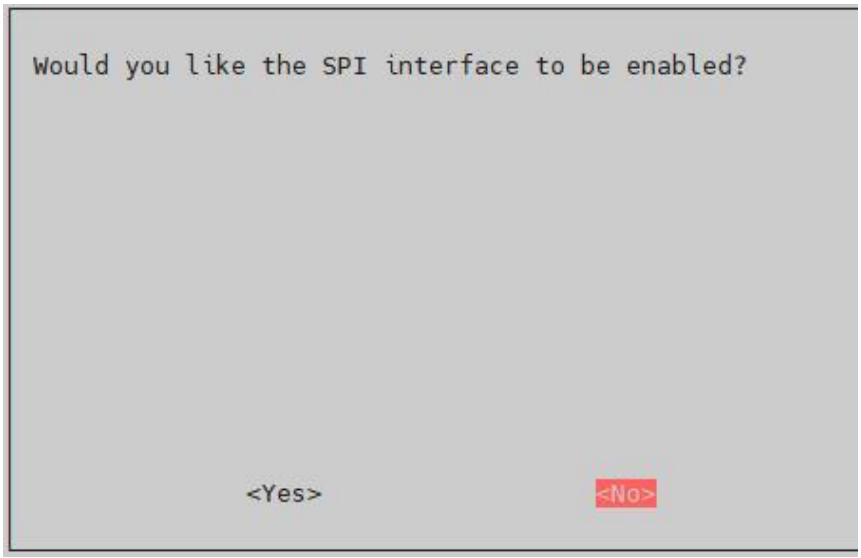
```
sudo raspi-config
Select Interface Options -> SPI -> Yes to disable the SPI interface
```



(/wiki/File:12.48-epaper-SPI-1.jpg)



(/wiki/File:12.48-epaper-SPI-2.jpg)



(/wiki/File:12.48-epaper-SPI-3.jpg)

- Restart Raspberry Pi:

```
sudo reboot
```

config.txt file setting

Use the command

```
sudo nano /boot/config.txt
```

or

```
sudo nano /boot/firmware/config.txt
```

Open the corresponding config.txt file

Add at the end

```
gpio=7=op,dl
gpio=8=op,dl
```

Press Ctrl+O (letter O) to save

Press Ctrl+X to exit

- Restart Raspberry Pi:

```
sudo reboot
```

Run C demo

- Install BCM2835 (Pi4, zero series, etc.)

```
#Open the Raspberry Pi terminal and run the following commands:
wget http://www.airspayce.com/mikem/bcm2835/bcm2835-1.71.tar.gz
tar zxvf bcm2835-1.71.tar.gz
cd bcm2835-1.71/
sudo ./configure && sudo make && sudo make check && sudo make install
# For more information, please refer to the official website: http://www.airspayce.com/mi
kem/bcm2835/
```

- Install wiringPi (Pi5)

```
git clone https://github.com/WiringPi/WiringPi
cd WiringPi
./build
gpio -v
# Run gpio -v and corresponding version will appear. If it does not appear, there is an i
nstallation error.
```

- Download the demo (you can skip it if you have already downloaded it)

```
https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_E.zip (http
s://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_E.zip)
unzip 13.3inch_e-Paper_E.zip -d 13.3inch_e-Paper_E
cd 13.3inch_e-Paper_E/RaspberryPi/
```

- Alternate decompression method [Expand]
- Download the demo via GitHub (alternative method, skip if already downloaded) [Expand]
- Compiler program (Note: -j4 means using 4 threads for compilation, the number can be modified by yourself)

```
# It is now in 13.3inch_e-Paper_E/RaspberryPi_JetsonNano
cd c
sudo make clean
# Use the BCM library to drive the Pi4
sudo make -j4 USELIB=USE_BCM2835_LIB
# Use the wiringPi library to drive the Pi5
sudo make -j4 USELIB=USE_WIRINGPI_LIB
```

- Run demo

```
sudo ./epd
```

Run python demo

- Install function library

```
sudo apt-get update
sudo apt-get install python3-pip
sudo apt-get install python3-pil
sudo apt-get install python3-numpy
sudo apt-get install python3-spidev
```

- Install function library (python2) [Expand]
- Download the demo (you can skip it if you have already downloaded it)

```
https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_E.zip (http
s://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_E.zip)
unzip 13.3inch_e-Paper_E.zip -d 13.3inch_e-Paper_E
cd 13.3inch_e-Paper_E/RaspberryPi/
```

- Alternate decompression method [Expand]
- Download the demo via GitHub (alternative method, skip if already downloaded) [Expand]
- Run demo

```
# Make sure it is in 13.3inch_e-Paper_E/RaspberryPi/
cd python/examples/
python3 epd_13in3e_test.py
```

Arduino

[Collapse]

Hardware connection

Use an 10PIN cable to connect, please refer to the pin corresponding table below:

Arduino connection pin correspondence

e-Paper	Arduino UNO	Mega2560
VCC	5V	5V
GND	GND	GND
DIN	D11	D51
CLK	D13	D52

CS_M	D10	D10
CS_S	D9	D9
DC	D8	D8
RST	D7	D7
BUSY	D6	D6
PWR	D5	D5

Install IDE

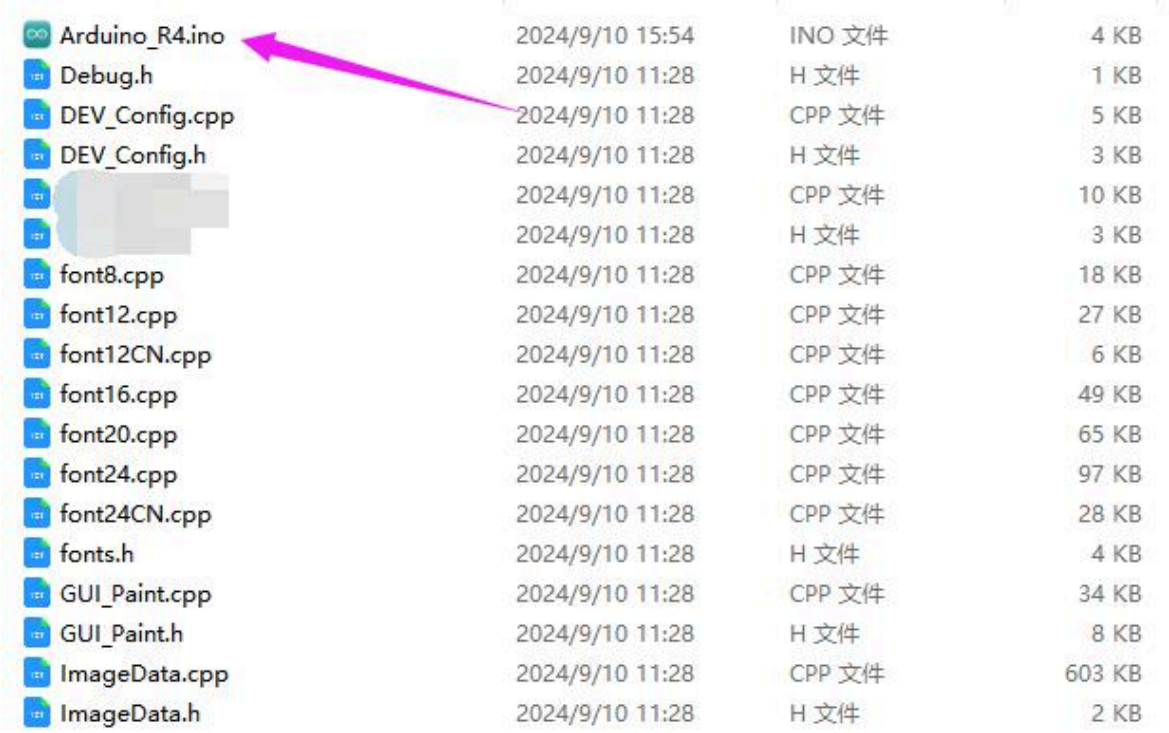
Arduino IDE Windows Installation Tutorial (https://www.waveshare.com/wiki/Arduino_ide_download)

Run demo

- In the Resources section download the Demo (https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_E.zip), then unzip it and enter the 13.3inch_e-Paper_E directory to see the following content

(/wiki/File:E-paper_arduino_cede-R4-2.jpg)

- Open the demo: 13.3inch_e-Paper_E\Arduino_R4\Arduino_R4.ino



Arduino_R4.ino	2024/9/10 15:54	INO 文件	4 KB
Debug.h	2024/9/10 11:28	H 文件	1 KB
DEV_Config.cpp	2024/9/10 11:28	CPP 文件	5 KB
DEV_Config.h	2024/9/10 11:28	H 文件	3 KB
	2024/9/10 11:28	CPP 文件	10 KB
	2024/9/10 11:28	H 文件	3 KB
font8.cpp	2024/9/10 11:28	CPP 文件	18 KB
font12.cpp	2024/9/10 11:28	CPP 文件	27 KB
font12CN.cpp	2024/9/10 11:28	CPP 文件	6 KB
font16.cpp	2024/9/10 11:28	CPP 文件	49 KB
font20.cpp	2024/9/10 11:28	CPP 文件	65 KB
font24.cpp	2024/9/10 11:28	CPP 文件	97 KB
font24CN.cpp	2024/9/10 11:28	CPP 文件	28 KB
fonts.h	2024/9/10 11:28	H 文件	4 KB
GUI_Paint.cpp	2024/9/10 11:28	CPP 文件	34 KB
GUI_Paint.h	2024/9/10 11:28	H 文件	8 KB
ImageData.cpp	2024/9/10 11:28	CPP 文件	603 KB
ImageData.h	2024/9/10 11:28	H 文件	2 KB

(/wiki/File:E-paper_arduino_cede-R4-3.jpg)

- In the Arduino IDE, select the corresponding development board (Board) and port (Port) in the Tools menu

(/wiki/File:E-paper_arduino_cede-R4-1.jpg)

- Finally, click Upload, and the upload is as follows (Arduino 1.8.13)

```
avrdude: AVR device initialized and ready to accept instructions

Reading | ################################################## | 100% 0.01s

avrdude: Device signature = 0x1e950f (probably m328p)
avrdude: reading input file
avrdude: writing flash (6916 bytes):

Writing | ################################################## | 100% 2.64s

avrdude: 6916 bytes of flash written

avrdude done.  Thank you.
```

(/wiki/File:E-paper_arduino_cede3.jpg)

STM32

[Collapse]

Hardware connection

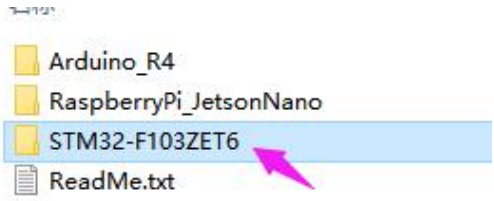
Use a 9PIN cable to connect, please refer to the pin corresponding table below:

STM32 connection pin correspondence

e-Paper	STM32
VCC	3.3V
GND	GND
DIN	PA7
CLK	PA5
CS_M	PA4
CS_S	PC4
DC	PA2
RST	PA1
BUSY	PA3
PWR	PA6

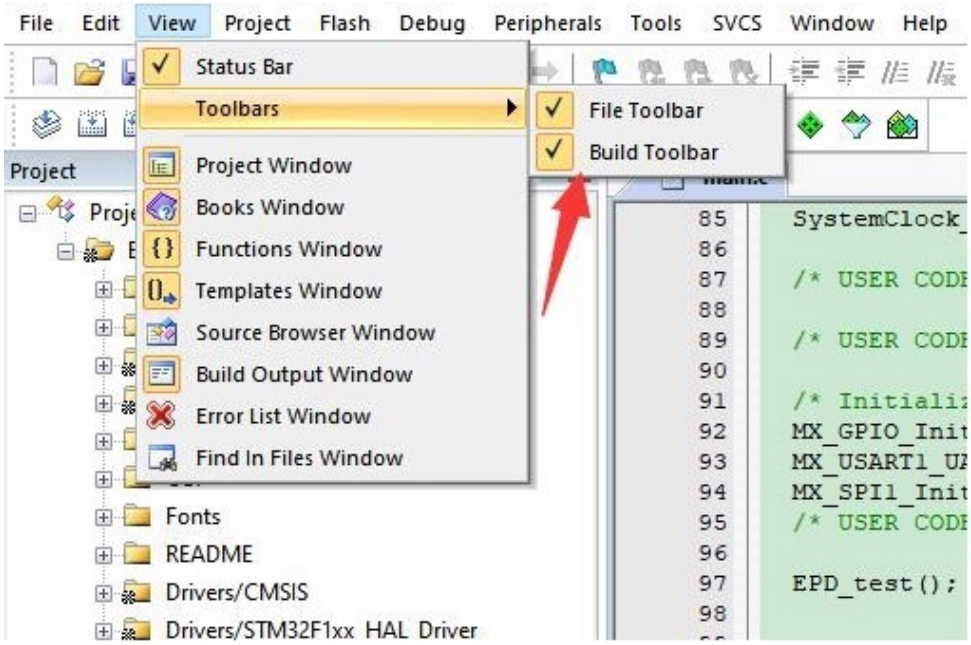
Run demo

- Click to download the Demo (https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_E.zip), then unzip it and enter the 13.3inch_e-Paper_E directory to see the following content



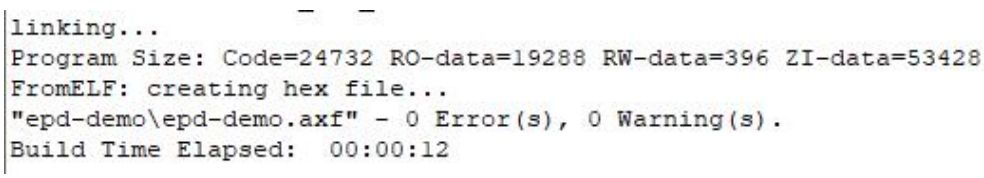
(/wiki/File:E-paper_stm32_cede1_f103zet6.jpg)

- Use Keil to open epd-demo.uvprojx in the 13.3inch_e-Paper_E\STM32-F103ZET6\MDK-ARM directory
- Open Keil's compilation toolbar (which is generally open) < br/>



(/wiki/File:E-paper_stm32_cede1.jpg)

- Click Compile



(/wiki/File:E-paper_stm32_cede3.jpg)

- Make sure the appropriate programmer is connected, then click LOAD to download the demo to the microcontroller

ESP32

[Collapse]

Hardware connection

ESP32 connection pin correspondence

e-Paper	ESP32
VCC	3.3V
GND	GND
DIN	IO14

CLK	IO13
CS_M	IO15
CS_S	IO2
DC	IO27
RST	IO26
BUSY	IO25
PWR	IO33

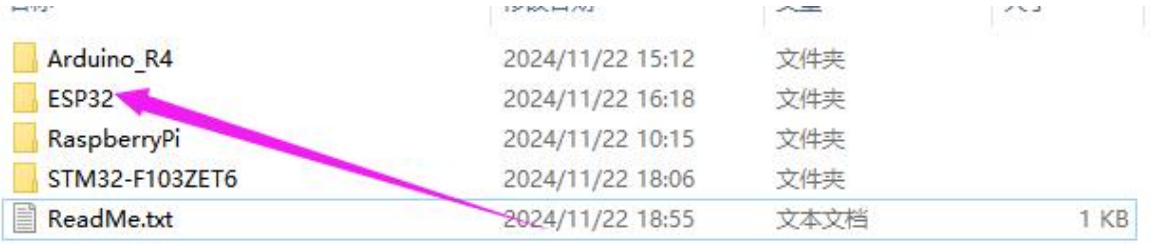
Environment configuration

Online installation requires the use of VPN software, please click the link below for more information.

- Install Online (https://www.waveshare.com/wiki/Arduino_ESP32/8266_Online_Installation)

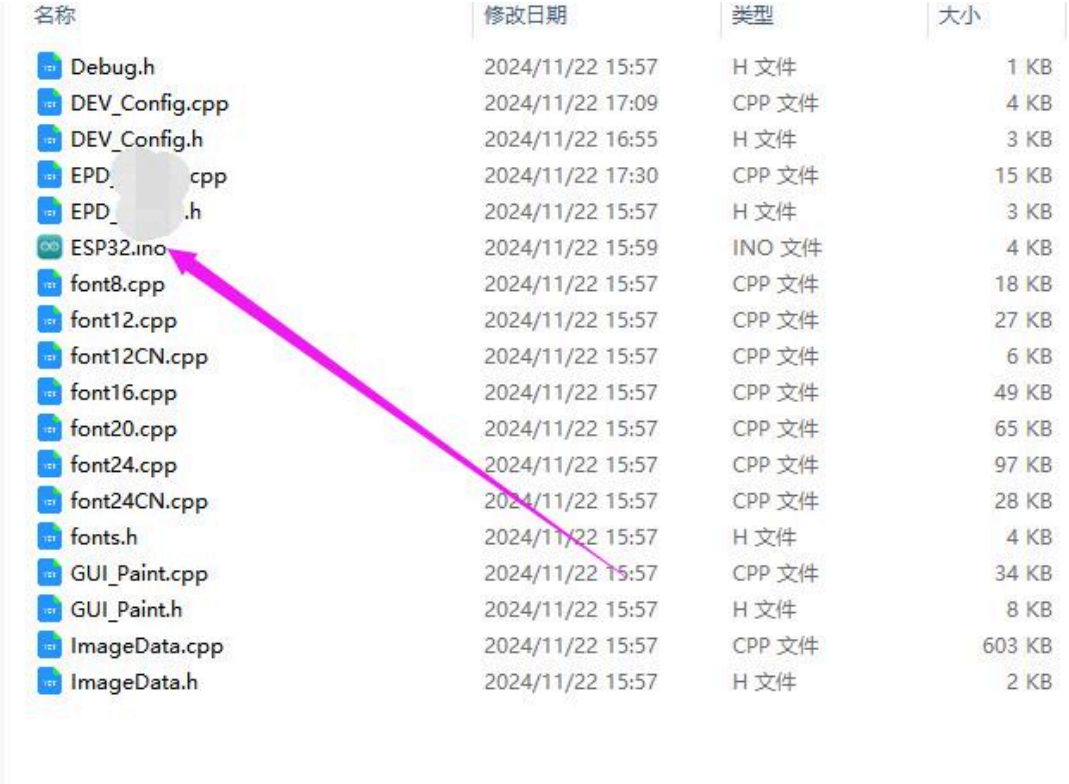
Run demo

- In the Resources section download the Demo (https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_E.zip), then unzip it and enter the 13.3inch_e-Paper_E directory to see the following content



(/wiki/File:Epd_esp32-1.jpg)

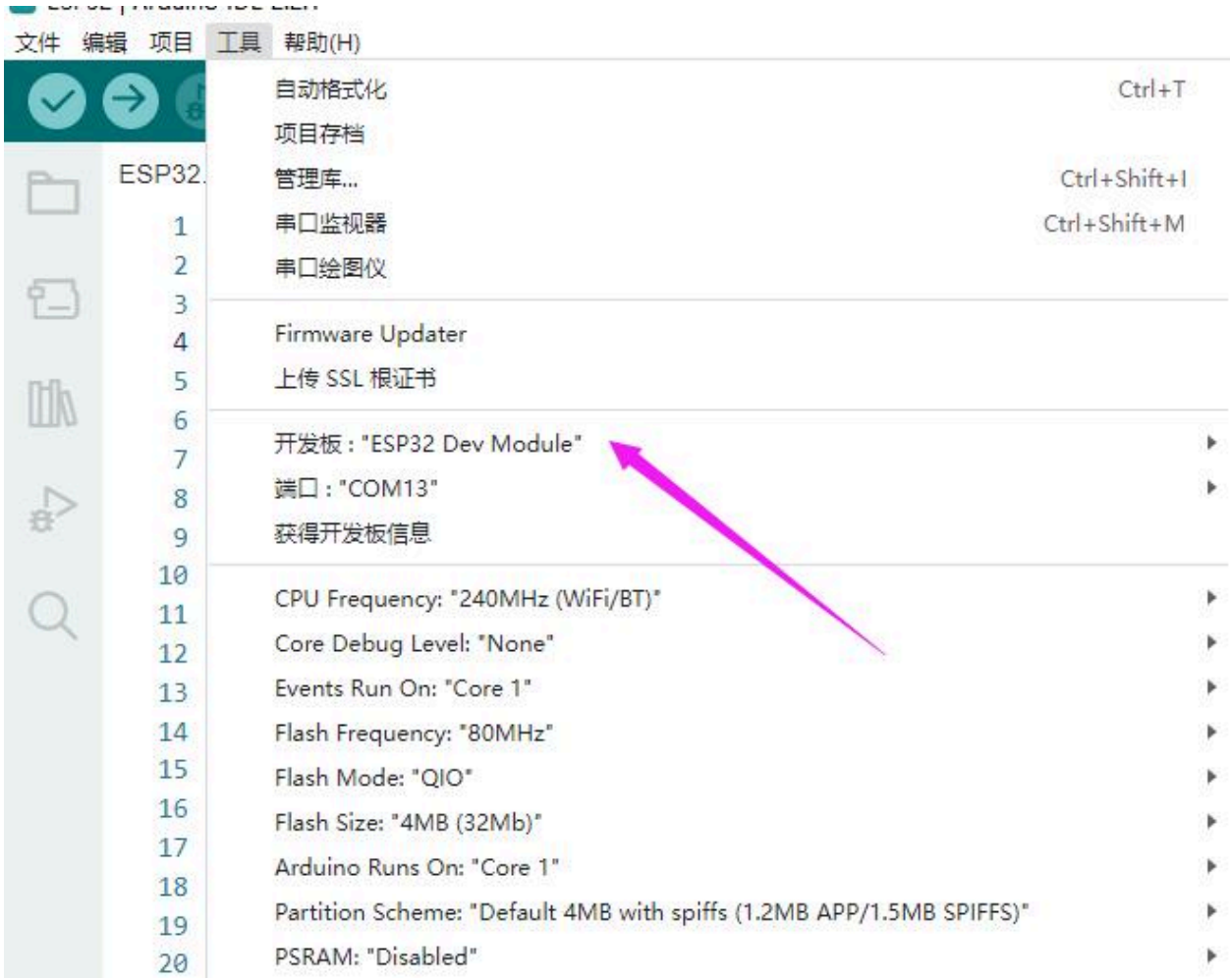
- Open the demo: 13.3inch_e-Paper_E\ESP32\ESP32.ino



名称	修改日期	类型	大小
Debug.h	2024/11/22 15:57	H 文件	1 KB
DEV_Config.cpp	2024/11/22 17:09	CPP 文件	4 KB
DEV_Config.h	2024/11/22 16:55	H 文件	3 KB
EPD_ cpp	2024/11/22 17:30	CPP 文件	15 KB
EPD_ .h	2024/11/22 15:57	H 文件	3 KB
ESP32.ino	2024/11/22 15:59	INO 文件	4 KB
font8.cpp	2024/11/22 15:57	CPP 文件	18 KB
font12.cpp	2024/11/22 15:57	CPP 文件	27 KB
font12CN.cpp	2024/11/22 15:57	CPP 文件	6 KB
font16.cpp	2024/11/22 15:57	CPP 文件	49 KB
font20.cpp	2024/11/22 15:57	CPP 文件	65 KB
font24.cpp	2024/11/22 15:57	CPP 文件	97 KB
font24CN.cpp	2024/11/22 15:57	CPP 文件	28 KB
fonts.h	2024/11/22 15:57	H 文件	4 KB
GUI_Paint.cpp	2024/11/22 15:57	CPP 文件	34 KB
GUI_Paint.h	2024/11/22 15:57	H 文件	8 KB
ImageData.cpp	2024/11/22 15:57	CPP 文件	603 KB
ImageData.h	2024/11/22 15:57	H 文件	2 KB

(/wiki/File:Epd_esp32-2.jpg)

- In the Arduino IDE, select the corresponding development board (Board) and port (Port) in the Tools menu



(/wiki/File:Epd_esp32-3.jpg)

- Finally, click Upload, and the upload is as follows (Arduino 2.2.1)

```
Writing at 0x000456f5... (53 %)
Writing at 0x0004aa83... (61 %)
Writing at 0x0004ff3d... (69 %)
Writing at 0x00055425... (76 %)
Writing at 0x0005e723... (84 %)
Writing at 0x00065b58... (92 %)
Writing at 0x0006b0de... (100 %)
Wrote 392192 bytes (210318 compressed) at 0x00010000 in 3.6 seconds (effective 882.0 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...
```

(/wiki/File:Epd_esp32-4.jpg)

ESP32-WiFi

please refer to https://www.waveshare.com/wiki/E-Paper_ESP32_Driver_Board#WiFi_Demo (https://www.waveshare.com/wiki/E-Paper_ESP32_Driver_Board#WiFi_Demo) program download： ESP32-wifi (https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/ESP32-wifi-13.3E.zip)

Resources

Documents

- Driver Board Schematic (https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_HAT%2B.pdf)
- 13.3inch e-Paper (E) Datasheet (https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_(E)_user_manual.pdf)

Demos

- Demo (https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/13.3inch_e-Paper_E.zip)
- ESP32-wifi (https://files.waveshare.com/wiki/13.3inch%20e-Paper%20HAT%2B/ESP32-wifi-13.3E.zip)

Development resources

- Create a Scatter Plot using Photoshop (https://www.waveshare.com/wiki/E-Paper_Floyd-Steinberg)
- E-Paper API Analysis (https://www.waveshare.com/wiki/E-Paper_API_Analysis)
- Character Conversion Software (https://files.waveshare.com/wiki/common/Zimo221.7z)
- E-Paper Font Library Conversion Tutorial (https://www.waveshare.com/wiki/Ink_Screen_Font_Library_Tutorial)
- Image2Lcd.7z Image Bitmap Conversion Software (https://files.waveshare.com/wiki/common/Image2Lcd.7z)

- Image2Lcd Image Bitmap Conversion Steps (https://www.waveshare.com/wiki/Image2Lcd_Image_Bitmap_Conversion)
- pwnagotchi Usage and Porting (https://www.waveshare.com/wiki/Template:2.13inch_e-Paper_HAT%2B_Pwn)

Related Links

- Raspberry Pi Documentation (https://www.waveshare.com/wiki/Raspberry_Pi_Documentation)
- e-Paper Dashboard (https://github.com/aceinnolab/Inkycal)
- Quectel-Pi e-Paper Reader (https://github.com/Quectel-Pi/demo-inkscreen-reader)
- GPS Tracker (https://github.com/mmolhoek/papertrail)
- TuyaOpen e-Paper Driver (https://github.com/tuya/TuyaOpen)

FAQ

[Collapse]

Software Issues

[Expand]

Hardware Issues

[Expand]

Screen Issues

[Expand]

Potential issues during development

[Expand]

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.
Working Time: 9 AM - 6 PM GMT+8 (Monday to Friday)

Submit Now (https://service.waveshare.com/)

Retrieved from "https://www.waveshare.com/w/index.php?title=13.3inch_e-Paper_HAT%2B_(E)_Manual&oldid=110104 (https://www.waveshare.com/w/index.php?title=13.3inch_e-Paper_HAT%2B_(E)_Manual&oldid=110104)"